# Trigonometry

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## Cosine and sine

**Definition 0.1** (Unit circle)**.** The **unit circle** is the set of points $(x_1, x_2)$ in the plane with $x_1^2 + x_2^2 = 1$.

**Example 0.1** (Points on the unit circle)**.**

- $(1, 0)$ is on the unit circle, because $1^2 + 0^2 = 1$.
- $(0, -1)$ is on the unit circle, because $0^2 + (-1)^2 = 1$.
- $(\frac{3}{5}, \frac{4}{5})$ is on the unit circle, because $\frac{9}{25} + \frac{16}{25} = 1$.
- $(1, 1)$ is not on the unit circle, because $1^2 + 1^2 = 2 \neq 1$.

**Definition 0.2** (The number $\pi$)**.** The number $\pi \approx 3.14159$ is half of the circumference of the unit circle (Definition 0.1).

**Example 0.2** (Distances around the unit circle)**.** The whole unit circle has circumference $2\pi$. Moving from $(1, 0)$ to $(-1, 0)$ along the upper half of the circle covers a distance of $\pi$, and moving from $(1, 0)$ to $(0, 1)$ covers a quarter of the circle, a distance of $\pi/2$.

**Definition 0.3** (Cosine)**.** For a real number $\theta$, start at the point $(1, 0)$ of the unit circle (Definition 0.1) and move a distance $|\theta|$ along the circle, counterclockwise if $\theta \geq 0$ and clockwise if $\theta < 0$. The first coordinate of the point reached is the **cosine** of $\theta$, written $\cos\theta$.

**Example 0.3** (Cosines of some angles)**.**

- $\theta = 0$ stays at $(1, 0)$, so $\cos 0 = 1$.

- $\theta = \pi$ goes half way around the circle, to $(-1, 0)$, so $\cos \pi = -1$.
- $\theta = \pi/2$ goes a quarter of the way round, to $(0, 1)$, so $\cos(\pi/2) = 0$.
- $\theta = \pi/4$ goes an eighth of the way round, to the point of the circle with $x_1 = x_2 > 0$. There $x_1^2 + x_1^2 = 1$, so $x_1^2 = \frac{1}{2}$ and $\cos(\pi/4) = x_1 = \frac{1}{\sqrt{2}} \approx 0.7071$.

Angles are also measured in *degrees*, with a full turn of $2\pi$ equal to 360 degrees, so $\pi/2$ is 90 degrees and $\pi/4$ is 45 degrees.

**Definition 0.4** (Sine)**.** For a real number $\theta$, the second coordinate of the point reached from $(1, 0)$ by moving a distance $|\theta|$ along the unit circle (Definition 0.3) is the **sine** of $\theta$, written $\sin \theta$.

**Example 0.4** (Sines of some angles)**.**

- $\theta = 0$ stays at $(1, 0)$, so $\sin 0 = 0$.
- $\theta = \pi$ goes half way around the circle, to $(-1, 0)$, so $\sin \pi = 0$.
- $\theta = \pi/2$ goes a quarter of the way round, to $(0, 1)$, so $\sin(\pi/2) = 1$.
- $\theta = \pi/4$ reaches the point with $x_1 = x_2 = \frac{1}{\sqrt{2}}$ (Example 0.3), so $\sin(\pi/4) = \frac{1}{\sqrt{2}}$.

**Theorem 0.1** (The Pythagorean identity)**.** *For every real number $\theta$,*

$$\cos^2 \theta + \sin^2 \theta = 1.$$

Proof

*Proof.* The point $(\cos \theta, \sin \theta)$ lies on the unit circle (Definition 0.1), so its coordinates satisfy $x_1^2 + x_2^2 = 1$. □

**Example 0.5** (The unit circle and the angle $\pi/3$)**.** Take $\theta = \pi/3$, one sixth of the way round the circle. Figure 1 shows the point reached, $(\cos \theta, \sin \theta)$, and its two coordinates. The point is a vertex of a regular hexagon inscribed in the unit circle, whose sides have the same length as its radius, so the point is at distance 1 from $(1, 0)$. Writing $(x_1, x_2) = (\cos \theta, \sin \theta)$,

$$\begin{aligned} 1 &= (x_1 - 1)^2 + x_2^2 \\ &= x_1^2 + x_2^2 - 2x_1 + 1 \\ &= 2 - 2x_1, \end{aligned}$$

so $\cos(\pi/3) = x_1 = \frac{1}{2}$, and then $\sin(\pi/3) = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$. To three decimal places these

are 0.5 and 0.866. As Theorem 0.1 says, the squared coordinates add to 1.

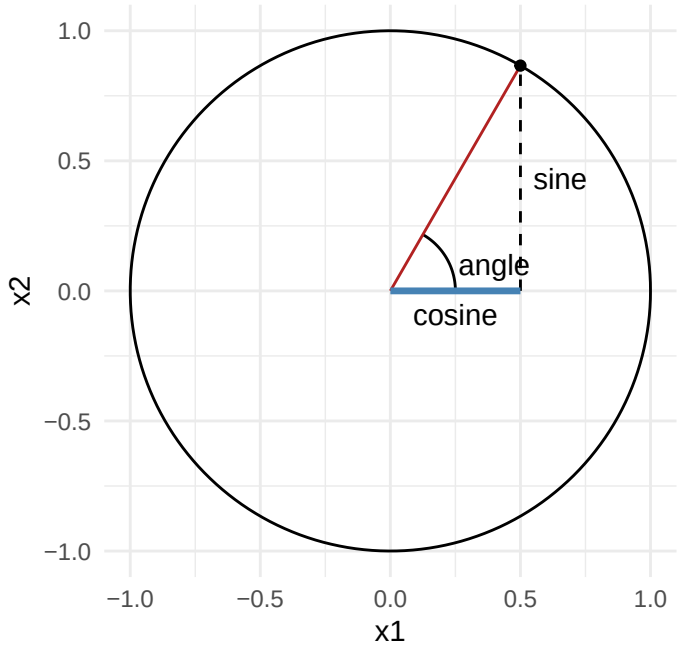


Figure 1: The unit circle with the point at angle $\pi/3$. Its horizontal coordinate is the cosine and its vertical coordinate is the sine.

**Theorem 0.2** (Cosine takes each value from $-1$ to 1 exactly once between 0 and $\pi$)**.** *The function* $\cos : [0, \pi] \to [-1, 1]$ *is decreasing, with* $\cos 0 = 1$ *and* $\cos \pi = -1$*, and takes each value in* $[-1, 1]$ *exactly once. On* $[0, \pi]$*,* $\sin \theta \geq 0$*.*

Proof

*Proof.* For $\theta \in [0, \pi]$, the point $(\cos\theta, \sin\theta)$ is at distance $\theta$ along the circle from $(1, 0)$, counterclockwise, so it lies on the upper half of the unit circle, $\{(x_1, x_2) : x_1^2 + x_2^2 = 1,\ x_2 \geq 0\}$, which gives $\sin\theta \geq 0$. Different values of $\theta$ in $[0, \pi]$ are different distances along that half circle, so they give different points. Two different points of the upper half have different first coordinates, because $x_2 = \sqrt{1 - x_1^2}$ is determined by $x_1$. So cos is one-to-one on $[0, \pi]$. Every $x_1 \in [-1, 1]$ is the first coordinate of the point $(x_1, \sqrt{1 - x_1^2})$ of the upper half, and the distance along the half circle from $(1, 0)$ to that point is some $\theta \in [0, \pi]$, so cos takes every value in $[-1, 1]$. Finally, move along the upper half circle from $(1, 0)$ to $(-1, 0)$ as $\theta$ increases from 0 to $\pi$. The first coordinate never takes the same value twice, so it never turns back. It starts at $\cos 0 = 1$

and ends at $\cos \pi = -1$, so it only decreases. $\square$